

Solutions — Problem Set 5

MACROECONOMIA 1 — MPE Insper — Prof. Pedro G. Duarte

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General Equilibrium — Kurlat (2020), ch. 9

How to read this document

Two questions, and the professor has built them as **mirror images**. Question 1 is chapter 9 with **capital frozen**: no investment technology, K fixed at $\bar{K}$, $\delta = 0$. Question 2 is chapter 9 with **labour frozen**: $L_t = 1$, no leisure in preferences. Between them they cover the whole chapter, one margin at a time. Neither is asking for the full three-agent model in one go, so each demands that you be very clean about the margin it left switched on.

The thread that joins them. In both, the object that has to be pinned down is a *price that nothing else determines*. In Question 1 the interest rate cannot move any goods — there is no storage and no investment — so r is purely the price that reconciles the household to a consumption path it cannot alter. In Question 2 the interest rate is tied to the marginal product of capital by arbitrage, and the steady state forces it back to pure impatience. *Preferences fix the interest rate; technology fixes the quantity of capital.*

Highlighted answers are in **blue**. **Red** boxes are traps; **green** boxes are method. Every number is reproduced by `Resolucao/lista5_codigo/`, which checks each analytical formula against an independent numerical route and aborts on any mismatch.

Language. Written in English, per the project's global output-language rule — and in any case the problem set itself is set in English.

Question	Points	Basis in Kurlat
1 — Two periods, fixed capital	5	§9.1–9.2; the machinery of Ex. 9.5 , 9.7
2 — Infinite horizon, fixed labour	5	§9.3; Ex. 9.11 , 9.12 ; (e) is Ex. 9.12 's punchline

Question 1 Two periods, capital fixed at $\bar{K}$

Read the amputation before writing anything

Three things are **gone** relative to Kurlat's §9.1, and each one removes a line you would otherwise be expected to write:

- **No investment firm.** There is no investment technology at all, so the third agent of ch. 9 does not exist and equation (9.1.11), $1 + r = r_2^K$, has no analogue.
- **No capital accumulation.** Equation (9.1.5) is replaced by $K_t = \bar{K}$ for both t . And $\delta = 0$.
- **No storage.** Therefore goods market clearing carries *no investment term*: $c_t = F(\bar{K}, L_t)$, period by period. This single line drives everything below.

What survives is exactly two margins: consumption against leisure *within* a period, and consumption today against consumption tomorrow *across* periods.

Item (a) — the two problems, and why $\Pi = 0$

The household.

$$\max_{c_1, l_1, c_2, l_2} u(c_1) + v(l_1) + \beta[u(c_2) + v(l_2)], \quad u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad v(l) = \psi \ln l,$$

subject to a budget constraint — and the constraint should be **built, not quoted**.

Building (1.1) out of the two period constraints.

The household trades a one-period bond a ($a > 0$ means it lends, $a < 0$ that it borrows), bought at date 1 and repaid with interest at date 2. Period by period, income must cover spending:

$$\text{date 1: } c_1 + a = w_1(1 - l_1) + r_1^K \bar{K} + \Pi_1, \quad \text{date 2: } c_2 = w_2(1 - l_2) + r_2^K \bar{K} + \Pi_2 + (1 + r)a.$$

Two remarks on the second one. The household ends with **nothing**: there is no date 3, so leaving assets behind would be strictly wasteful (u is increasing), and leaving debt behind is not permitted. And $\bar{K}$ appears only through its *rent* at both dates, never as a principal — see the red box below.

Now divide the date-2 constraint by $1 + r$ and add it to the date-1 one. The a terms are $+a$ and $-\frac{(1+r)a}{1+r} = -a$, so they **cancel**:

$$c_1 + \frac{c_2}{1+r} = w_1(1 - l_1) + \frac{w_2(1 - l_2)}{1+r} + r_1^K \bar{K} + \frac{r_2^K \bar{K}}{1+r} + \Pi, \quad \Pi \equiv \Pi_1 + \frac{\Pi_2}{1+r}. \quad (1.1)$$

Where the bond went — and why item (b) has only one λ

a vanished from (1.1), and that is the whole content of “there is a credit market”. Borrowing is not a separate decision to be optimised: once c_1 and l_1 are chosen, the date-1 constraint *defines* what a must be,

$$a = w_1(1 - l_1) + r_1^K \bar{K} + \Pi_1 - c_1,$$

and since a never enters the utility function, the household cares about the consumption path and is indifferent to how it is financed. **Two constraints became one, which is exactly why there will be one multiplier and not two** in item (b) — and why the Euler equation exists at all. Without credit, $a \equiv 0$ would be forced, the two constraints would stay separate, and the household would be stuck consuming its income date by date.

In *equilibrium* $a = 0$ anyway — a representative household has nobody to trade with — but that is a conclusion, not an assumption. The household is free to borrow; r moves until it does not want to. See item (c).

The one place this differs from Kurlat (9.1.1)

Kurlat’s household holds $\bar{K}(1 + r_1^K - \delta)$, which includes the **principal** because in his model capital can be sold into the used-capital market. **Here it cannot.** There is no investment technology, so no market exists in which a unit of capital could be turned back into consumption goods. The household can only *rent* it out, and it does so in both periods. Hence rental income only, $r_1^K \bar{K}$ and $r_2^K \bar{K}$, and no $(1 - \delta)$ term anywhere. Say this explicitly in your answer — writing Kurlat’s term here imports a market this economy does not have.

The firm, separately in each period $t = 1, 2$.

$$\Pi_t^F = \max_{K_t, L_t} A_t K_t^\alpha L_t^{1-\alpha} - w_t L_t - r_t^K K_t \quad (1.2)$$

The firm is **static**: it rents everything period by period and carries nothing between dates, so it has no discount factor and no intertemporal problem.

Why profit is zero.

Because F has **constant returns to scale** and factors are paid their marginal products. The Cobb–Douglas exponents sum to one, so F is homogeneous of degree one and Euler’s theorem gives

$$F(K, L) = K F_K + L F_L.$$

Verify it directly: $F_K = \alpha F/K$ and $F_L = (1 - \alpha)F/L$, so $KF_K + LF_L = \alpha F + (1 - \alpha)F = F$. ✓
Now impose the firm's first-order conditions $F_K = r_t^K$ and $F_L = w_t$:

$$r_t^K K_t + w_t L_t = \alpha F + (1 - \alpha)F = F(K_t, L_t) \quad \implies \quad \Pi_t^F = 0.$$

Revenue exactly exhausts factor payments. ✓ (checked in `l5q1.py::check_zero_lucro`).

Economic reading. This is not an equilibrium condition that has to be imposed — it is an *identity* that follows from constant returns plus competitive pricing. Its role is to let Π drop out of (1.1), so that in equilibrium household wealth is just labour income plus rental income. Note also that it is *scale* that delivers zero profit, not α : any constant-returns technology would do.

Item (b) — the first-order conditions

Step 1 — the Lagrangian.

Write the right-hand side of (1.1) as **wealth**,

$$W(l_1, l_2) \equiv w_1(1 - l_1) + \frac{w_2(1 - l_2)}{1 + r} + r_1^K \bar{K} + \frac{r_2^K \bar{K}}{1 + r} + \Pi,$$

keeping the arguments visible: an hour of leisure at date t costs w_t of wealth, and that is the only reason a wage will appear in the leisure condition below. The constraint holds with **equality** because u is strictly increasing — the household never ends up holding unspent wealth — so we may attach a single multiplier λ to it:

$$\mathcal{L} = \underbrace{u(c_1) + v(l_1) + \beta[u(c_2) + v(l_2)]}_{\text{measured in utils}} + \lambda \underbrace{\left[W(l_1, l_2) - c_1 - \frac{c_2}{1+r} \right]}_{\text{measured in date-1 goods}}.$$

The two braces are in **different units**. λ is the exchange rate between them, and everything below is bookkeeping.

Step 2 — differentiate.

Four choice variables, four conditions:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_1} &= u'(c_1) - \lambda = 0, & \frac{\partial \mathcal{L}}{\partial c_2} &= \beta u'(c_2) - \frac{\lambda}{1+r} = 0, \\ \frac{\partial \mathcal{L}}{\partial l_1} &= v'(l_1) - \lambda w_1 = 0, & \frac{\partial \mathcal{L}}{\partial l_2} &= \beta v'(l_2) - \frac{\lambda w_2}{1+r} = 0, \end{aligned}$$

that is,

$$\underbrace{u'(c_1) = \lambda}_{\text{(i)}}, \quad \underbrace{\beta u'(c_2) = \frac{\lambda}{1+r}}_{\text{(ii)}}, \quad \underbrace{v'(l_1) = \lambda w_1}_{\text{(iii)}}, \quad \underbrace{\beta v'(l_2) = \frac{\lambda w_2}{1+r}}_{\text{(iv)}}.$$

Every one of the four reads the same sentence: *marginal utility of the thing* = $\lambda \times$ *what one unit of that thing costs, priced in date-1 goods*. A unit of c_1 costs 1; a unit of c_2 costs $\frac{1}{1+r}$, because that is what must be set aside today to have one good tomorrow; an hour of leisure costs the wage w_t , discounted if it falls at date 2.

Notice the asymmetry that will produce the Euler equation: β sits on the **utility** side and is the household's own impatience, while $\frac{1}{1+r}$ sits on the **price** side and is the market's. Optimality is the point where the two are reconciled.

What λ is — three readings of the same number

1. Mechanically, a unit converter. The objective is in utils, the constraint in date-1 goods. You cannot set one equal to the other, so the first-order conditions need a conversion factor. λ carries the units *utils per date-1 good*.

2. Economically, the marginal utility of wealth — the shadow price of the budget constraint. Let $V(W)$ be maximised lifetime utility as a function of wealth. The envelope theorem gives

$$\lambda = V'(W),$$

so λ answers the question: *hand this household one more unit of date-1 goods — by how much does its lifetime utility rise?* A poor household has a large λ , a rich one a small λ .

3. Concretely, here it equals $u'(c_1)$. Condition (i) says precisely that:

$$\lambda = u'(c_1) = c_1^{-\sigma} > 0.$$

Positive, which confirms the constraint binds. And it is no accident: the budget was written in **date-1 goods**, and the household can always turn one unit of date-1 wealth into one unit of c_1 , so the value of wealth *must* come out equal to the marginal utility of present consumption. Write the same constraint in date-2 units and the multiplier comes out as $\beta u'(c_2)$ instead. The number moves with the numeraire; the economics does not. **λ is a price, not a preference parameter.**

Counting. Five unknowns ($c_1, c_2, l_1, l_2, \lambda$) against five equations (the four conditions plus the constraint itself). λ is determined — it is simply not needed for the two boxed results, because both are ratios.

Why there is *one* λ and not two — the credit market is hiding in here

There is one multiplier because there is one constraint, and there is one constraint because the household **can borrow and lend at r** . That is exactly what licenses collapsing the two period budgets into the single present-value constraint (1.1).

Suppose credit were unavailable, so the household had to satisfy

$$c_1 \leq w_1(1 - l_1) + r_1^K \bar{K} \quad \text{and} \quad c_2 \leq w_2(1 - l_2) + r_2^K \bar{K}$$

separately. Then there would be two multipliers, λ_1 and λ_2 , the consumption conditions would read $u'(c_1) = \lambda_1$ and $\beta u'(c_2) = \lambda_2$, and dividing one by the other would give

$$\frac{u'(c_1)}{\beta u'(c_2)} = \frac{\lambda_1}{\lambda_2} \neq 1 + r \quad \text{in general.}$$

No Euler equation. The Euler equation is not a law of nature — it is the footprint of a market. Seeing that the *same* λ appears at both dates *is* seeing that the household can move goods through time at the price $1 + r$. (This is the same point as the credit-limit exercise in ch. 6: bind the household and the Euler equation becomes an inequality.)

Step 3 — take ratios, and λ cancels twice.

λ is one number, not one per date, so any ratio of two of the four conditions is free of it. That is why it never has to be computed, and why the content of optimality is always a *ratio*: a marginal rate of substitution set equal to a relative price.

The *intratemporal* margin — consumption against leisure.

Dividing the leisure condition by the consumption condition, in each period separately — (iii)/(i) at date 1, and (iv)/(ii) at date 2, where β and $\frac{1}{1+r}$ cancel along with λ ,

$$\frac{v'(l_1)}{u'(c_1)} = \frac{\lambda w_1}{\lambda} = w_1, \quad \frac{\beta v'(l_2)}{\beta u'(c_2)} = \frac{\lambda w_2/(1+r)}{\lambda/(1+r)} = w_2,$$

which is why the same condition holds in both periods and carries no intertemporal price:

$$\boxed{\frac{v'(l_t)}{u'(c_t)} = w_t} \iff \boxed{\frac{\psi c_t^\sigma}{1 - L_t} = w_t}, \quad t = 1, 2. \quad (1.3)$$

This is Kurlat (9.1.7). **It is the condition that describes the trade-off between consumption and leisure.**

The *intertemporal* margin — present against future consumption.

Dividing the c_1 condition by the c_2 condition, (i)/(ii):

$$\frac{u'(c_1)}{\beta u'(c_2)} = \frac{\lambda}{\lambda/(1+r)} = 1 + r,$$

which rearranges to

$$\boxed{u'(c_1) = \beta(1+r) u'(c_2)} \iff \boxed{\left(\frac{c_2}{c_1}\right)^\sigma = \beta(1+r)} \quad (1.4)$$

This is the **Euler equation**, Kurlat (9.1.8). **It is the condition that describes the trade-off between present and future consumption.**

The firm.

$$\boxed{w_t = (1 - \alpha) A_t \bar{K}^\alpha L_t^{1-\alpha} = (1 - \alpha) \frac{Y_t}{L_t}}, \quad \boxed{r_t^K = \alpha A_t \bar{K}^{\alpha-1} L_t^{1-\alpha} = \alpha \frac{Y_t}{\bar{K}}} \quad (1.5)$$

σ or $1/\sigma$?

State it once, explicitly, and you cannot be marked wrong under either convention: σ is the **curvature of u** (the coefficient of relative risk aversion), and $1/\sigma$ is the **elasticity of intertemporal substitution**. It matters in item (e), where the sign of the whole answer turns on $1 - \sigma$.

Item (c) — the definition of competitive equilibrium

A **competitive equilibrium** is an allocation $\{c_1, c_2, l_1, l_2, L_1, L_2, K_1, K_2\}$ and a price system $\{r, w_1, w_2, r_1^K, r_2^K\}$ such that:

1. **Household optimality.** Given the prices, the allocation solves the utility maximisation problem subject to the present-value budget constraint (1.1) — that is, it satisfies (1.3) and (1.4).
2. **Firm optimality.** Given the prices, in each period the firm's choice of (K_t, L_t) solves (1.2) — that is, it satisfies (1.5).
3. **Market clearing**, in each period:

$$\text{goods: } \boxed{c_t = A_t \bar{K}^\alpha L_t^{1-\alpha}}, \quad \text{labour: } \boxed{L_t + l_t = 1}, \quad \text{capital: } \boxed{K_t = \bar{K}}. \quad (1.6)$$

Why the credit market clears without being listed.

It is worth proving rather than asserting. Take the household's *date-1* constraint, the one that still contains the bond, and impose what we already know: $L_1 = 1 - l_1$ from labour clearing, $\Pi_1 = 0$ from item (a), and Euler's theorem $w_1 L_1 + r_1^K \bar{K} = F(\bar{K}, L_1) = Y_1$. Then

$$c_1 + a = w_1 L_1 + r_1^K \bar{K} + \Pi_1 = Y_1.$$

Goods clearing says $c_1 = Y_1$. Subtracting,

$$a = 0$$

Three markets clearing force the fourth. That is **Walras's law**, and it is why the definition above need not list the credit market: doing so would be redundant, not wrong. With a representative household there is also a one-line version of the same conclusion — *the representative agent cannot trade with itself* — but the accounting argument is the one that generalises.

Where the marks are

Three things earn the marks here, and all three are easy to drop.

- **All three conditions.** A student who writes only the first-order conditions has given *demand*, not equilibrium.
- **Goods clearing has no investment term.** $c_t = Y_t$, period by period. This is where the amputation shows up, and every result in (d) and (e) descends from it.
- **The credit market.** Net lending is zero, $a = 0$. Say that it follows from the others by **Walras's law**, and that with a representative household it must hold anyway: *the representative agent cannot trade with itself*.

Item (d) — equilibrium labour, and the interest rate

Step 1: substitute the firm's condition into the household's.

Put $w_t = F_L$ from (1.5) into (1.3):

$$\underbrace{\frac{v'(l_t)}{u'(c_t)}}_{\text{MRS}_{l,c}} = \underbrace{F_L(\bar{K}, L_t)}_{\text{MRT}_{l,c}} \quad (1.7)$$

which is exactly the form the question asks for, and is Kurlat (9.1.12). On the left, the rate at which the household *is willing* to trade leisure for consumption. On the right, the rate at which the economy *is able* to convert leisure into consumption — which is the marginal product of labour. The wage has cancelled: it appeared once in a household condition and once in a firm condition.

Step 2: substitute goods market clearing.

Using $c_t = A_t \bar{K}^\alpha L_t^{1-\alpha}$ and $v'(l_t) = \psi/(1 - L_t)$, $u'(c_t) = c_t^{-\sigma}$:

$$\frac{\psi}{1 - L_t} \left(A_t \bar{K}^\alpha L_t^{1-\alpha} \right)^\sigma = (1 - \alpha) A_t \bar{K}^\alpha L_t^{-\alpha}.$$

Take the power through on the left, separating the L_t from the rest:

$$\frac{\psi (A_t \bar{K}^\alpha)^\sigma L_t^{(1-\alpha)\sigma}}{1 - L_t} = (1 - \alpha) A_t \bar{K}^\alpha L_t^{-\alpha}.$$

Now multiply both sides by L_t^α , which clears the negative exponent on the right and merges the two powers of L_t on the left into $L_t^{(1-\alpha)\sigma+\alpha}$; then divide by $\psi (A_t \bar{K}^\alpha)^\sigma$, which leaves $(A_t \bar{K}^\alpha)^{1-\sigma}$ on the right:

$$\frac{L_t^{\alpha+(1-\alpha)\sigma}}{1 - L_t} = \frac{1 - \alpha}{\psi} (A_t \bar{K}^\alpha)^{1-\sigma} \quad (1.8)$$

Why (1.8) has exactly one solution — say this

The left-hand side is the product of $L^{\alpha+(1-\alpha)\sigma}$, strictly increasing from 0, and $1/(1-L)$, strictly increasing to $+\infty$. So it rises **strictly monotonically from 0 to $+\infty$** on $(0,1)$, while the right-hand side is a positive constant. Exactly one crossing. ✓ (verified on a grid of 2×10^5 points in `l5q1.py::check_unicidade`; panel (i) of Figure 1 draws it.)

There is **no closed form**, and none is expected — Kurlat's own key to PSET 4 stops at an implicit equation for exactly this reason. Characterise it and differentiate implicitly.

Note that (1.8) involves only date- t objects. **The two periods do not interact on the labour margin at all**, precisely because no goods can be moved between them.

Q1(d) — the equation that pins down L_t ($\alpha = 0.35$, $\psi = 1.8$, $\bar{K} = 2$)

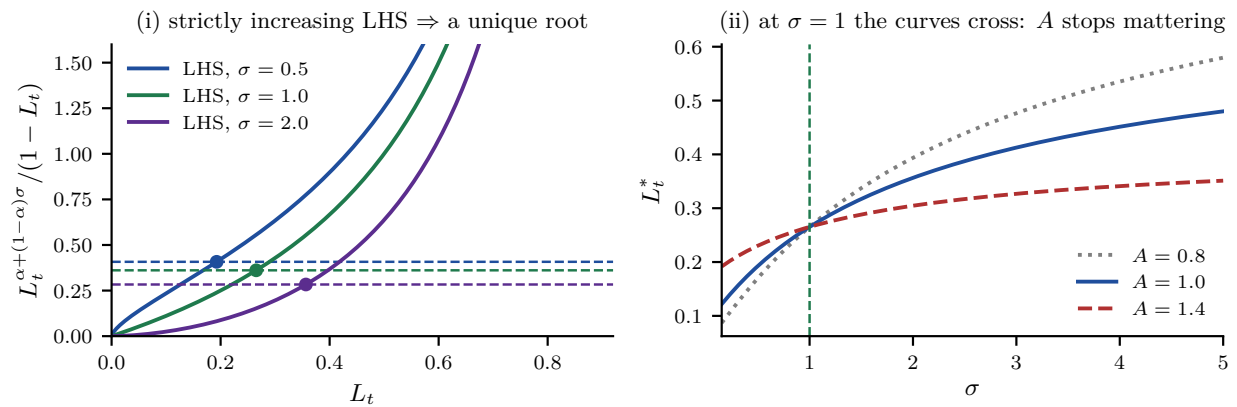


Figure 1: Q1(d). Left: the strictly increasing left-hand side of (1.8) against the constant right-hand side — one crossing, for every σ . Right: equilibrium hours against σ ; at $\sigma = 1$ the three curves meet, because productivity drops out of (1.8).

Step 3: the interest rate.

From the Euler equation (1.4) with goods clearing (1.6):

$$1 + r = \frac{1}{\beta} \left(\frac{c_2}{c_1} \right)^\sigma, \quad \frac{c_2}{c_1} = \frac{A_2}{A_1} \left(\frac{L_2}{L_1} \right)^{1-\alpha},$$

so

$$1 + r = \frac{1}{\beta} \left[\frac{A_2}{A_1} \left(\frac{L_2}{L_1} \right)^{1-\alpha} \right]^\sigma \quad (1.9)$$

with L_1 and L_2 each solving (1.8) at their own A_t . ✓

Economic reading. Read what (1.9) does *not* contain: no marginal product of capital. In Kurlat's full model the interest rate is tied to F_K by arbitrage, because capital can be accumulated. Here it cannot, so r has no technological anchor at all. It is a pure intertemporal price whose only job is to make the household *content* with a consumption path it is powerless to change — c_t is forced to equal Y_t by (1.6). The interest rate here allocates nothing; it merely reports.

The special case worth stating.

If $A_2 = A_1$ then (1.8) is the same equation in both periods, so $L_2 = L_1$, so $c_2 = c_1$, and

$$1 + r = \frac{1}{\beta} \quad \text{i.e.} \quad r = \rho \equiv \frac{1}{\beta} - 1.$$

With $\beta = 0.96$ that is $r = 4.17\%$. ✓ *When there is nothing to trade and no growth, the interest rate is exactly impatience.*

Item (e) — the elasticity of hours to productivity**Step 1: take logs of (1.8).**

$$[\alpha + (1 - \alpha)\sigma] \ln L_t - \ln(1 - L_t) = \ln \frac{1 - \alpha}{\psi} + (1 - \sigma)(\ln A_t + \alpha \ln \bar{K}).$$

Step 2: differentiate with respect to $\ln A_t$.The right-hand side contributes $1 - \sigma$. On the left, the second term needs the chain rule:

$$\frac{d \ln(1 - L_t)}{d \ln A_t} = \frac{1}{1 - L_t} \cdot \left(-L_t \frac{d \ln L_t}{d \ln A_t} \right) = -\frac{L_t}{1 - L_t} \frac{d \ln L_t}{d \ln A_t},$$

and it enters with a minus sign in front, so it *adds*. Collecting:

$$\frac{d \ln L_t}{d \ln A_t} \left[\alpha + (1 - \alpha)\sigma + \frac{L_t}{1 - L_t} \right] = 1 - \sigma.$$

Step 3: divide.

$$\boxed{\frac{d \ln L_t}{d \ln A_t} = \frac{1 - \sigma}{\alpha + (1 - \alpha)\sigma + \frac{L_t}{1 - L_t}}} \quad (1.10)$$

as required. ✓ Checked in `15q1.py` against a symmetric finite difference at $h = 10^{-6}$, for five values of σ between 0.25 and 4.**Step 4: sign it.**The denominator is a sum of three strictly positive terms ($\alpha > 0$; $(1 - \alpha)\sigma > 0$; and $L_t/(1 - L_t) > 0$ since $L_t \in (0, 1)$). Therefore

$$\text{sign} \left(\frac{d \ln L_t}{d \ln A_t} \right) = \text{sign}(1 - \sigma).$$

$$\boxed{\text{Higher productivity raises hours if and only if } \sigma < 1.}$$

At $\sigma = 1$ (log utility in consumption) hours do not move at all; for $\sigma > 1$ they fall.**Step 5: sign the three prices and quantities.**Write $\varepsilon_L \equiv d \ln L_t / d \ln A_t$ for the elasticity just found, and take logs of (1.5) and of goods clearing (1.6):

$$\ln w_t = \ln(1 - \alpha) + \ln A_t + \alpha \ln \bar{K} - \alpha \ln L_t,$$

$$\ln r_t^K = \ln \alpha + \ln A_t - (1 - \alpha) \ln \bar{K} + (1 - \alpha) \ln L_t, \quad \ln c_t = \ln A_t + \alpha \ln \bar{K} + (1 - \alpha) \ln L_t.$$

 $\bar{K}$ is a constant and drops out of every derivative. Differentiating each with respect to $\ln A_t$ — the direct term gives 1, the $\ln L_t$ term gives its coefficient times ε_L —

$$\frac{d \ln w_t}{d \ln A_t} = 1 - \alpha \varepsilon_L, \quad \frac{d \ln r_t^K}{d \ln A_t} = \frac{d \ln c_t}{d \ln A_t} = 1 + (1 - \alpha) \varepsilon_L. \quad (1.11)$$

The last two coincide because $\ln r_t^K$ and $\ln c_t$ differ only by a constant — read on.

All three rise, for every σ — and here is why

The wage. $\alpha\varepsilon_L < 1 - \sigma \leq 1$ when $\sigma < 1$, since the denominator of (1.10) exceeds α ; and when $\sigma > 1$ the term $-\alpha\varepsilon_L$ is positive. Either way $d \ln w / d \ln A > 0$. ✓

The rental rate and consumption. The only worry is $\sigma > 1$, where $\varepsilon_L < 0$. But $\alpha + (1 - \alpha)\sigma + \frac{L}{1-L} - (1 - \alpha)(\sigma - 1) = 1 + \frac{L}{1-L} = \frac{1}{1-L} > 0$, so $(1 - \alpha)|\varepsilon_L| < 1$ always. ✓

And note the exact identity $r_t^K = \alpha c_t / \bar{K}$: with $\bar{K}$ fixed and goods clearing forcing $c_t = Y_t$, the rental rate is consumption times a constant. That is why the last two elasticities in (1.11) are not merely similar but *identical*. ✓

The numbers.

Calibration $\alpha = 0,35$, $\psi = 1,8$, $\bar{K} = 2$, $A = 1$:

σ	L_t^*	$d \ln L / d \ln A$	$d \ln w / d \ln A$	$d \ln r^K / d \ln A = d \ln c / d \ln A$
0,25	0,14423	+1,1013	+0,6146	+1,7158
0,50	0,19273	+0,5472	+0,8085	+1,3557
1,00	0,26531	0	+1	+1
2,00	0,35647	−0,4537	+1,1588	+0,7051
4,00	0,45136	−0,7952	+1,2783	+0,4831

✓ Every row verified by finite differences on the *re-solved* equilibrium.

Trace the pattern across the rows: as σ rises, equilibrium hours rise, the hours response to productivity turns from positive to negative, the wage response strengthens past one, and the consumption response weakens below one. Every one of those movements is the income effect gaining ground on the substitution effect.

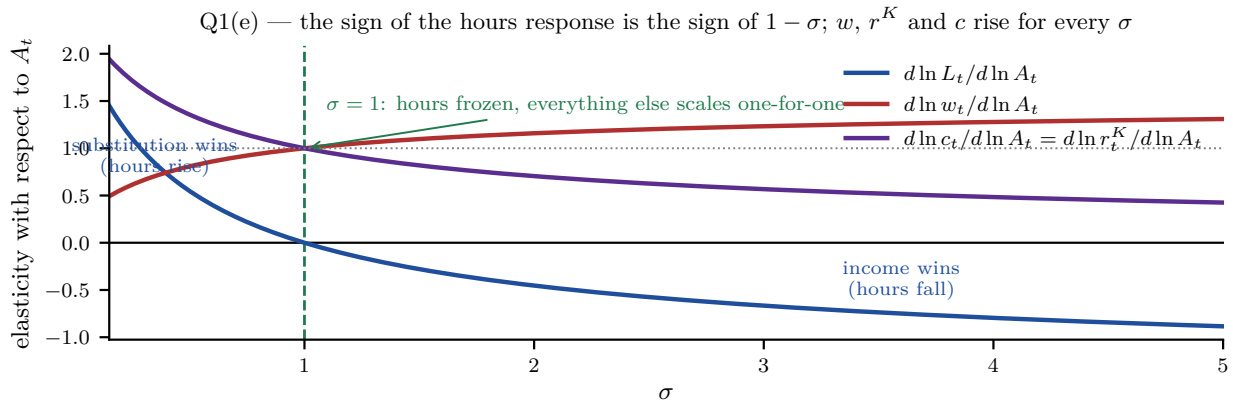


Figure 2: Q1(e). The sign of the hours response is the sign of $1 - \sigma$; $\sigma = 1$ is the knife edge, and there everything else scales one-for-one with A .

Income and substitution effects.

A rise in A_t raises the marginal product of labour and therefore the wage. That does two things, and they pull opposite ways.

- **Substitution effect.** Leisure is now more expensive — an hour of leisure costs more consumption than before — so the household substitutes away from leisure and *works more*.
- **Income effect.** At any given number of hours the household is richer. Leisure is a normal good, so it wants more of it and *works less*.

σ is the arbiter. σ is how fast the marginal utility of consumption falls as consumption rises. If σ is **large**, marginal utility collapses quickly, so the extra consumption the higher wage would buy is not worth much: the household takes the gain as leisure, the income effect dominates, and

hours fall. If σ is **small**, marginal utility falls slowly, the extra consumption is still worth having, the substitution effect dominates, and hours rise.

Where the marks are

At $\sigma = 1$ *exactly* the two cancel. That is not a coincidence: logarithmic utility in consumption is the unique case in which income and substitution effects on labour supply offset one another exactly, which is why it is the standard assumption in growth models that want hours to be stationary along a balanced growth path.

You have already seen this knife edge. In Problem Set 4, with $\ln c$ and no transfer, hours were completely independent of the wage and of the tax — the effect had to be carried by the lump-sum transfer T . Here the same cancellation is what makes $\sigma = 1$ the dividing line. *Same fact, seen twice.*

Question 2 Infinite horizon, labour fixed at $L_t = 1$

Item (a) — the household problem and the Euler equation

$$\max_{\{c_t, a_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\sigma}}{1-\sigma},$$

subject to the flow budget constraint in every period

$$a_{t+1} = (1 + r_t)a_t + w_t + \Pi_t - c_t, \quad (2.1)$$

with a_0 given by the initial capital stock K_0 , and a **No-Ponzi condition**

$$\lim_{T \rightarrow \infty} \frac{a_{T+1}}{\prod_{t=0}^T (1 + r_t)} \geq 0. \quad (2.2)$$

Labour income is w_t because $L_t = 1$, and $\Pi_t = 0$ by the constant-returns argument of Question 1(a). **Say explicitly why (2.2) is needed:** without it the household would roll debt over forever, financing unbounded consumption, and the problem would have no solution.

The Euler equation, by a perturbation.

Cut c_t by ε and save the proceeds. Utility cost today: $u'(c_t)\varepsilon$. The ε becomes $(1 + r_{t+1})\varepsilon$ tomorrow, worth $\beta(1 + r_{t+1})u'(c_{t+1})\varepsilon$ in today's units. At an optimum no such perturbation can help:

$$\boxed{u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1})} \iff \boxed{\frac{c_{t+1}}{c_t} = [\beta(1 + r_{t+1})]^{1/\sigma}} \quad (2.3)$$

Economic reading. The Euler equation is a no-arbitrage condition on the household's *own* consumption plan. The left side is the marginal cost of giving up a unit today; the right side is the marginal benefit, discounted because tomorrow matters less.

The same result with multipliers — and what λ_t is here.

The perturbation argument is the fastest route, but it is worth seeing the Lagrangian once, because the multiplier changes character relative to Question 1. There is now **one constraint per date**, so there is one multiplier per date:

$$\mathcal{L} = \sum_{t=0}^{\infty} \left\{ \beta^t u(c_t) + \lambda_t [(1 + r_t)a_t + w_t + \Pi_t - c_t - a_{t+1}] \right\}$$

Two first-order conditions. In c_t , and in a_{t+1} — which appears with a minus sign in the date- t constraint and inside $(1 + r_{t+1})a_{t+1}$ in the date- $(t + 1)$ one:

$$\frac{\partial \mathcal{L}}{\partial c_t} : \quad \beta^t u'(c_t) = \lambda_t, \quad \frac{\partial \mathcal{L}}{\partial a_{t+1}} : \quad \lambda_t = \lambda_{t+1}(1 + r_{t+1}).$$

Substituting the first into the second, $\beta^t u'(c_t) = \beta^{t+1} u'(c_{t+1})(1 + r_{t+1})$, and dividing by β^t returns (2.3) exactly. ✓

Here $\lambda_t = \beta^t u'(c_t)$ is the **discounted** marginal utility of date- t consumption: the value, measured in date-0 utils, of handing the household one more good at date t . The second condition says $\lambda_t/\lambda_{t+1} = 1 + r_{t+1}$ — the multipliers fall through time at exactly the interest rate, because that is the rate at which goods can be moved. In Question 1 the two dates had already been merged into a single constraint, so a single λ did this job; the ratio $\lambda_t/\lambda_0 = 1/\prod_{s \leq t} (1 + r_s)$ is the same object as the $\frac{1}{1+r}$ that sat inside (1.1). *Same economics, two bookkeeping conventions.*

The condition that closes the problem.

The Euler equation is necessary, not sufficient: it fixes the *slope* of the consumption path and says nothing about its *level*. What pins the level down is the **transversality condition**

$$\lim_{T \rightarrow \infty} \beta^T u'(c_T) a_{T+1} = 0. \quad (2.3')$$

Read it together with No-Ponzi (2.2): the household may not die in debt (2.2), and it must not die holding assets of positive present value either, since consuming them would raise utility at no cost (2.3'). Between them: **spend everything, eventually**. This is the infinite-horizon counterpart of the step in Question 1(a) where the date-2 constraint was imposed with equality because there was no date 3.

What makes consumption grow.

From (2.3), $c_{t+1} > c_t$ if and only if $\beta(1 + r_{t+1}) > 1$, that is

$$r_{t+1} > \frac{1}{\beta} - 1 \equiv \rho$$

Consumption grows exactly when the market's reward for waiting exceeds the household's impatience. When it does not, the path slopes down.

The role of σ .

It enters only through the exponent $1/\sigma$, which is the **elasticity of intertemporal substitution**. It governs how strongly the household tilts its path for a *given* gap between r and ρ . High σ means low elasticity: the household strongly resists letting consumption differ across dates, so even a large gap produces only a gentle slope.

What σ does *not* do

σ affects only **how fast** the economy travels, never **where it ends up**. It vanishes from the steady-state condition in item (d) — setting $c_{t+1}/c_t = 1$ raises the bracket to the power $1/\sigma$, and $1^{1/\sigma} = 1$ for any σ . ✓ (verified for $\sigma \in \{0.5; 1; 2; 3; 8\}$ in `15q2.py::check_estado_estacionario`.)

Item (b) — the firm's problem and factor prices

$$\max_{K_t, L_t} AK_t^\alpha L_t^{1-\alpha} - w_t L_t - r_t^K K_t$$

First-order conditions, evaluated at $L_t = 1$:

$$\boxed{w_t = (1 - \alpha)AK_t^\alpha}, \quad \boxed{r_t^K = \alpha AK_t^{\alpha-1}} \quad (2.4)$$

The wage is *increasing* in the capital stock and the rental rate *decreasing* in it — both by diminishing returns: more capital per worker raises the marginal product of labour and lowers the marginal product of capital. Profits are zero, so household income is wages plus capital income and nothing else.

Item (c) — who owns the capital, and why it does not matter

Arrangement (i): the household owns K , rents it out, and invests itself.

Its budget constraint is

$$c_t + I_t = w_t + r_t^K K_t, \quad K_{t+1} = (1 - \delta)K_t + I_t.$$

Substitute $I_t = K_{t+1} - (1 - \delta)K_t$ out and collect the K_t terms:

$$c_t + K_{t+1} = w_t + \underbrace{(r_t^K + 1 - \delta)}_{\text{gross return on capital}} K_t. \quad (2.5)$$

The bracket is the point: a unit of capital carried into date t pays its rent r_t^K and survives as $1 - \delta$ units of machine, so its gross return is the sum of the two. That is already the whole of (2.7) in embryo — a bond carried into date t pays $1 + r_t$, and nothing distinguishes the two but the label.

Now derive the condition instead of quoting it. Solve (2.5) for c_t ,

$$c_t = w_t + (r_t^K + 1 - \delta)K_t - K_{t+1},$$

and substitute into the objective, so that the household is choosing the *sequence* $\{K_{t+1}\}$ directly. Each K_{t+1} then appears in exactly **two** terms of $\sum_s \beta^s u(c_s)$: at date t , where raising it costs consumption one-for-one, and at date $t + 1$, where it pays the gross return. Hence

$$\frac{\partial}{\partial K_{t+1}} \sum_s \beta^s u(c_s) = \beta^t u'(c_t) \cdot (-1) + \beta^{t+1} u'(c_{t+1})(r_{t+1}^K + 1 - \delta) = 0,$$

and dividing through by β^t :

$$-u'(c_t) + \beta u'(c_{t+1})(r_{t+1}^K + 1 - \delta) = 0 \implies u'(c_t) = \beta(r_{t+1}^K + 1 - \delta)u'(c_{t+1}). \quad (2.6)$$

If the household holds bonds *as well*, its constraint is $c_t + a_{t+1} + K_{t+1} = w_t + (1 + r_t)a_t + (r_t^K + 1 - \delta)K_t$, and the same exercise in a_{t+1} returns (2.3). Both assets are held in equilibrium, so both must offer the same return — otherwise the household would take an unbounded long position in one and short the other, and no equilibrium would exist. Comparing (2.6) with the bond Euler equation (2.3), the marginal utilities cancel and:

$$\boxed{1 + r_{t+1} = r_{t+1}^K + 1 - \delta} \quad (2.7)$$

Arrangement (ii): a separate investment firm.

Its profit is Kurlat (9.3.3):

$$\Pi_t^I = \max_I \frac{r_{t+1}^K + 1 - \delta}{1 + r_{t+1}} [(1 - \delta)K_t + I] - [(1 - \delta)K_t + I].$$

Differentiate:

$$\frac{\partial \Pi_t^I}{\partial I} = \frac{r_{t+1}^K + 1 - \delta}{1 + r_{t+1}} - 1,$$

which is a **constant**: the problem is *linear* in I and has no interior optimum. Three cases, and only one survives:

Case	What the firm does	Consistent with equilibrium?
ratio > 1	$I \rightarrow +\infty$, infinite profit	No — infinite capital
ratio < 1	$I \rightarrow -\infty$	No
ratio $= 1$	any I , profit exactly zero	Yes

Setting the ratio to one gives (2.7) again. **The identical condition.** ✓ (both routes, plus the linearity claim itself, in `l5q2.py::check_arbitragem`).

The subtlety almost everyone gets wrong

The investment firm's first-order condition **does not determine how much is invested**. It determines a **price**. At the no-arbitrage return the firm is genuinely indifferent across every level of I , so the quantity of investment must come from somewhere else — namely **goods market clearing**, item (d). Looking for “the investment firm's optimal I ” is looking for something that does not exist.

Why the allocation does not depend on who owns the capital.

Three reasons; give all three.

- 1. The same intertemporal price.** Both arrangements deliver (2.7), so the household faces the same trade-off between a unit of consumption today and one tomorrow, whether it holds the claim as physical capital or as a bond. The rate at which the *economy* can move goods through time is a technological fact; the no-arbitrage condition just makes the financial price match it.
- 2. Zero profits in both.** The productive firm earns zero by constant returns; the investment firm earns zero because that is precisely what (2.7) enforces. No arrangement transfers any surplus to anybody, so the household's lifetime budget set is *identical*.
- 3. Nothing real has moved.** Consumption possibilities depend only on the economy's resource constraint and on the intertemporal price, and ownership touches neither.

Economic reading. This is **Fisher separation** (Fisher, *The Theory of Interest*, 1930), and it has the same flavour as Modigliani–Miller: with complete and competitive markets, how a claim on a real asset is packaged does not change real outcomes. In one sentence: the investment firm is an *accounting device*, not an agent with independent influence, because arbitrage strips it of any profit and hands the entire intertemporal trade-off back to the household unchanged.

Item (d) — resource constraint, steady state, closed forms

The resource constraint.

Goods market clearing with $L_t = 1$ is $AK_t^\alpha = c_t + I_t$; capital accumulation is $K_{t+1} = (1 - \delta)K_t + I_t$. Substituting I_t out:

$$K_{t+1} = (1 - \delta)K_t + AK_t^\alpha - c_t \quad (2.8)$$

In words: the capital you carry into tomorrow is what survives depreciation, plus everything you produced, minus everything you ate. Together with (2.3) this is a pair of difference equations in (K_t, c_t) , and from here everything is geometry.

The steady state.

Constant consumption in (2.3) requires $\beta(1+r) = 1$, so $1+r = 1/\beta$. Combining with the arbitrage condition (2.7) and the firm's condition (2.4), $r = r^K - \delta = F_K(K, 1) - \delta$:

$$F_K(K^*, 1) - \delta = \frac{1}{\beta} - 1 \quad (2.9)$$

which is what the question asks you to show, and is Kurlat (9.3.17).

Closed forms.

With $F_K = \alpha AK^{\alpha-1}$:

$$\alpha A(K^*)^{\alpha-1} = \frac{1}{\beta} - 1 + \delta \quad \Rightarrow \quad K^* = \left[\frac{\alpha A}{\frac{1}{\beta} - 1 + \delta} \right]^{\frac{1}{1-\alpha}} = \left[\frac{\alpha A \beta}{1 - \beta + \beta \delta} \right]^{\frac{1}{1-\alpha}} \quad (2.10)$$

and setting $K_{t+1} = K_t$ in (2.8):

$$c^* = A(K^*)^\alpha - \delta K^* \quad (2.11)$$

— output minus replacement investment. In a steady state *all* investment is replacement investment.

✓ Both verified as a genuine rest point of (2.3) and (2.8), for four values of β and five of σ .

The steady-state interest rate.

$$r^* = \frac{1}{\beta} - 1 = \rho$$

It depends on β and on nothing else — not on A , not on α , not on δ .

Why — and this is the punchline of the question

In a steady state consumption is constant, and the Euler equation says a constant path requires $\beta(1+r) = 1$ *exactly*. That is a statement purely about **preferences**: the reward for waiting must precisely offset impatience, which is the only configuration under which the household is content to consume the same amount forever. Technology has no say in it.

What technology determines is the **quantity** of capital. Given that r must equal $1/\beta - 1$, the capital stock adjusts until $F_K - \delta$ matches that number. A more productive economy does not end up with a higher interest rate; it ends up with **more capital**, which drives F_K back down until it meets what preferences demand.

Preferences determine the interest rate; technology determines the capital stock. Say it in exactly those words.

The numbers.

$A = 1$, $\alpha = 0,35$, $\delta = 0,08$:

β	K^*	c^*	$F_K - \delta$	r^*	K^*/K_{gr}
0,900	2,53676	1,18221	0,11111	11,11%	0,262
0,960	5,08153	1,35991	0,04167	4,17%	0,525
0,990	8,06630	1,43122	0,01010	1,01%	0,833
0,999	9,50193	1,43889	0,00100	0,10%	0,981

Moving β from 0,90 to 0,99 — a change in impatience of about ten percentage points a year — **more than triples** the steady-state capital stock. Patience is enormously powerful for capital. Consumption, however, moves only from 1,18 to 1,43, about 21%. Item (e) explains that asymmetry.

$$Q2(d) — \beta = 0.96, \sigma = 2, \alpha = 0.35, \delta = 0.08, A = 1$$

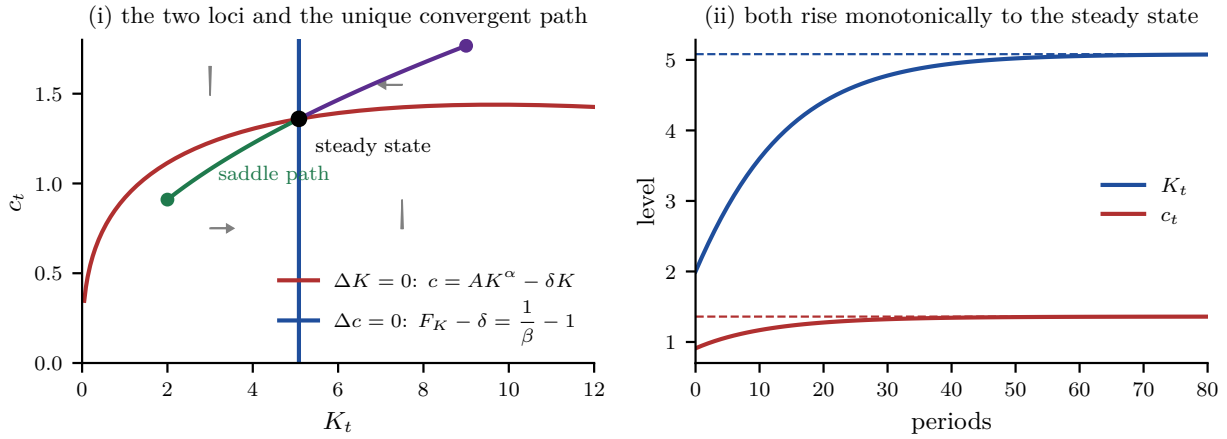


Figure 3: Q2(d). Left: the $\Delta c = 0$ locus is *vertical* because c does not appear in (2.9); the $\Delta K = 0$ locus is the concave hump $c = AK^\alpha - \delta K$. Only one trajectory converges — the saddle path. Right: the same transition against time. K is a *state* variable and cannot jump; c is a *jump* variable and must.

Item (e) — Bonus: the Golden Rule

Derivation.

The Golden Rule maximises steady-state consumption. From (2.11), maximise $AK^\alpha - \delta K$ over K :

$$\alpha AK^{\alpha-1} - \delta = 0 \quad \Rightarrow \quad \boxed{F_K(K_{gr}, 1) = \delta}, \quad \boxed{K_{gr} = \left(\frac{\alpha A}{\delta}\right)^{\frac{1}{1-\alpha}}} \quad (2.12)$$

✓ (confirmed as the argmax of $c^*(K)$ on a grid of 6×10^5 points).

The comparison.

Put the two conditions side by side:

$$\text{Golden Rule: } F_K = \delta \quad \text{vs} \quad \text{competitive steady state: } F_K = \delta + \underbrace{\left(\frac{1}{\beta} - 1\right)}_{> 0 \text{ iff } \beta < 1}.$$

Since $\beta < 1$, the marginal product of capital at the competitive steady state is **strictly greater** than at the Golden Rule. And F_K is strictly *decreasing* in K . Therefore

$$\boxed{K^* < K_{gr}}$$

With the calibration above, $K_{gr} = 9,685$ and $c_{gr} = 1,43898$, so at $\beta = 0,96$ the economy holds only 52% of the Golden Rule capital stock. ✓

Why an efficient economy stops short.

The First Welfare Theorem says the competitive equilibrium is Pareto efficient. So why not accumulate “enough”?

1. **Reject the premise hidden in “enough”.** The Golden Rule maximises consumption *in the steady state*. It says nothing about the path taken to get there and gives no weight at all to *when* consumption arrives. It is a criterion about one point, not about a trajectory.
2. **Say what the household actually maximises.** The household maximises the *discounted* sum of utility over the whole path. Moving from K^* up to K_{gr} requires investing more — consuming *less* — for a long stretch, in exchange for higher consumption in the far future. An impatient household discounts that far-future gain and ranks the whole package *below* where it already is.
3. **Conclude sharply. Under-accumulation relative to the Golden Rule is not a market failure. It is the optimum.** There is no Pareto improvement available — which is exactly what the First Welfare Theorem asserts — and the reason there is none is that reaching the Golden Rule would demand a sacrifice today that the household does not want to make. **The Golden Rule is a benchmark, not a target;** it stops being normative the moment somebody in the model has preferences.

The contrast that makes the point

In the **Solow** model with an *exogenous* saving rate the economy can perfectly well end up on the *other* side, with $K > K_{gr}$. That is **dynamic inefficiency**, and it is a genuine failure: consumption could be raised in every period, starting immediately, simply by saving less. Kurlat’s Exercise 6.7 (p. 124) constructs exactly that case, with $s = 0,4$ against $\alpha = 0,35$, producing a negative real interest rate.

What the present model shows is that dynamic inefficiency **can never happen** once the saving rate is *chosen* rather than assumed. An optimising household would never over-accumulate. Only an exogenous saving rate can produce that pathology.

The limit as $\beta \rightarrow 1$.

As $\beta \rightarrow 1$, $\frac{1}{\beta} - 1 \rightarrow 0$, so (2.9) converges to $F_K = \delta$, which is precisely (2.12). Hence

$$\beta \rightarrow 1 \implies K^* \rightarrow K_{gr}$$

✓ At $\beta = 1 - 10^{-9}$ the ratio K^*/K_{gr} is 1 to six decimal places. **The distance between the competitive steady state and the Golden Rule is impatience, measured.**

Q2(e) — $K^{ss} < K_{gr}$, and the two converge as $\beta \rightarrow 1$

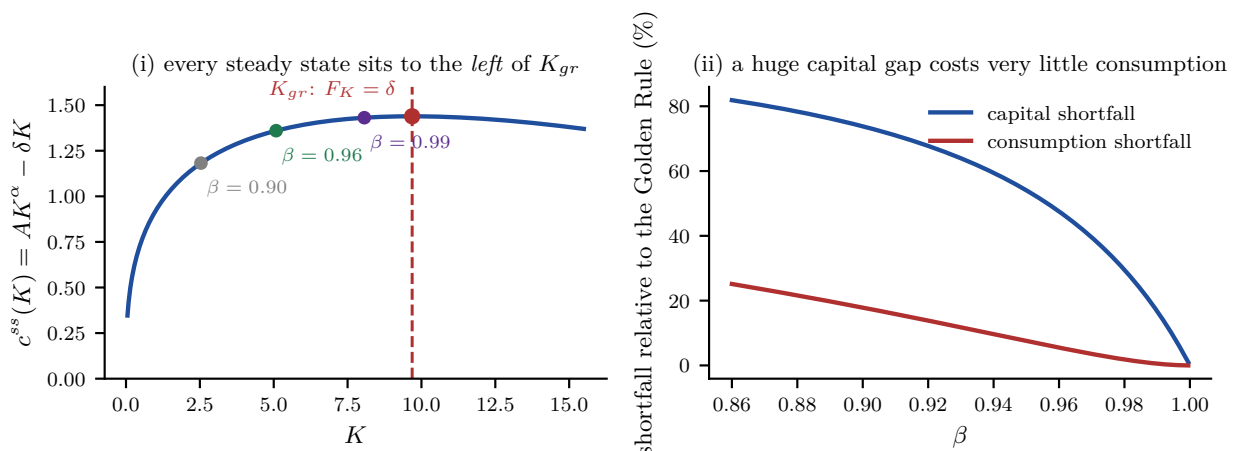


Figure 4: Q2(e). Left: every steady state lies to the left of K_{gr} , and slides toward it as β rises. Right: the two shortfalls. Because $c^*(K)$ is at a maximum at K_{gr} , its first derivative there is zero and the consumption loss is *second order*.

One observation worth adding, and most students miss it

At $\beta = 0,999$ the capital stock is 1,9% short of the Golden Rule, but consumption is short by only **0,006%**. Even at $\beta = 0,96$, where capital is barely half the Golden Rule level, consumption is only 5,5% below.

The reason is that K_{gr} is a *maximum*, so $c^*(K)$ is flat there: the first derivative vanishes and the loss from being off the peak is **second order**. Two consequences. First, arguments that an economy is losing a lot of consumption by under-accumulating are usually quantitatively weak. Second, this is the same flatness that makes Kurlat's Exercise 9.12 (p. 185) come out as it does, where optimising the entire savings path for 150 years is worth about a third of one percent of consumption relative to using the correct constant rate.

Companion code

Every number above is reproduced by `Resolucao/lista5_codigo/`. Each script checks its analytical formulas against an independent numerical route and **aborts on any mismatch**.

File	What it does
<code>modelo.py</code>	Every closed form, once: Q1's two equivalent forms of the labour equation, the elasticities, the interest rate; Q2's steady state, Golden Rule, the two difference equations, and a bisection that shoots for the saddle path.
<code>15q1.py</code>	Zero profit via Euler's theorem; the raw and tidied labour equations share the same unique root for 5 values of σ and 4 of A ; the LHS of (1.8) is strictly increasing on a 2×10^5 -point grid; (1.10) against a symmetric finite difference, and the induced elasticities of w , r^K and c against their own; $r^K = \alpha c / \bar{K}$ exactly; and the interest rate, including the $A_2 = A_1$ collapse to $1/\beta - 1$.
<code>15q2.py</code>	Both ownership routes give the same r , and the investment firm's profit is identically zero in I at that r and strictly monotone at any other; the steady state is a rest point for every σ ; $K^* < K_{gr}$; K_{gr} is the argmax of c^* ; $K^* \rightarrow K_{gr}$ as $\beta \rightarrow 1$; and the saddle path converges while neighbouring paths fall off it in both directions.
<code>15_figuras.py</code>	The four figures.
<code>estilo_mpl.py</code>	<code>pgf</code> backend, so figure text is typeset by <code>pdflatex</code> with <code>lmodern</code> and <code>cmap</code> and the PDF stays fully copyable.